

Rupsha Das

gmail | portfolio | linkedin | github

Experience

Full-Stack Intern, ZedBlox

Hyderabad | Aug 2025 – Jul 2026

- Built and deployed an **LLM-powered Telemetry Analytics Assistant** using **OpenAI GPT** and **Google Gemini** for shipment diagnostics, battery health analysis, anomaly detection, and telemetry insights.
- Developed full-stack features for an IoT management platform supporting **1,000+ connected medical devices**, including React interfaces, REST APIs, role-based access control, OTP-protected OTA workflows, metadata versioning, and centralized activity logging.
- Engineered backend services using **Node.js**, **Express.js**, **MongoDB Aggregation Framework**, and **REST APIs**; applied indexed queries, server-side downsampling, and parallel query execution to **reduce API payloads by 65%**, **improve query performance by 92%**, and **reduce dashboard latency by 40%**.
- Worked across Admin Console and Customer Console applications by redesigning data models, telemetry dashboards, and reusable React components to unify monitoring across the **ActiPod** and **SafePod** ecosystems while delivering scalable, responsive, and user-focused product experiences.
- Optimized stream-based CSV/PDF reporting and export pipelines through reusable backend services, **reducing report generation time by 50%** and lowering memory usage.

Embedded Systems Intern, IEEE Computational Intelligence Society Kolkata

June 2025 – July 2025

- Built an autonomous **edge-AI vehicle** on the **ESP32-S3** using a quantized **YOLOv5n** model for real-time object detection and on-device navigation.
- Developed embedded firmware integrating **computer vision**, **motor control**, **sensor processing**, and **wireless communication** under strict memory and compute constraints.
- Implemented sensor fusion using ultrasonic sensors and IR wheel encoders to improve obstacle avoidance, localization, and motion accuracy.

Projects

SophistAI – Personal Syllabus Navigator

sophistai.app

- Developed an **LLM-powered learning platform** that processes syllabus PDFs, converts unstructured academic content into structured knowledge maps, topic hierarchies, prerequisite relationships, and personalized learning paths, and generates actionable study plans using **natural language processing and prompt engineering**.
- Secured **1st place among 200+ teams** at the Diversion 2k25 Hackathon and attracted **500+ users**.

Manimate – Agentic AI Education Platform

GitHub

- Built an **agentic AI education platform** using **Next.js** that transforms academic topics into structured lessons, narrated explanations, and high-fidelity **Manim** videos through automated lesson planning and Python animation-code generation.
- Designed an end-to-end pipeline combining web research, multi-provider LLM generation, code execution, traceback-aware self-correction, Manim rendering, Kokoro TTS, FFmpeg video stitching, and AI-generated quizzes.

AnimeScope – Anime Music Player and Radio Experience

Live Demo | GitHub

- Developed an anime-inspired music player using **JavaScript**, **TypeScript**, **CSS**, **Node.js**, and **Express.js**, featuring play/pause, shuffle, repeat, seek, volume controls, searchable playlists, a local catalog without API keys, optional YouTube Data API search, animated snow/rain/sakura effects, and JSON APIs for songs, playlists, wallpapers, quotes, and listener counts.

Education

University of Kalyani Kalyani

2022 – 2026

B.Tech in Computer Science and Engineering

DGPA: 7.36

Technical Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, HTML, CSS, **Rust**

Frontend: React.js, Next.js, Redux, Tailwind CSS, Framer Motion, Radix UI, Responsive UI, Reusable Component Design

Backend: Node.js, Express.js, Django, Flask, REST APIs, Authentication, Role-Based Access Control, API Integration, MongoDB, MySQL, **SQL Server**, **Redis**, **PostgreSQL**

Systems & Databases: MongoDB Aggregation Framework, Query Optimization, Data Modeling, SQL, Indexing, Pagination, Server-Side Downsampling, Stream Processing, **Distributed Systems**, **System Design**, **Scalability**, **Performance Optimization**, **0-5 years experience**

AI & ML: LLM Application Development, Prompt Engineering, OpenAI, Gemini, Anthropic, Mistral, TensorFlow Lite, Computer Vision, YOLOv5, Anomaly Detection, Edge AI, Retrieval Workflows

Cloud & Tools: AWS S3, AWS EC2, **Azure**, Docker, MongoDB Atlas, Git, GitHub, Postman, Vercel, Netlify, Render, Firebase, FFmpeg, Manim, **Serverless**

Platforms: Linux, Windows, Web, ESP32-S3, **Consumer Platforms**, **Serverless**

Professional Skills: **Problem Solving**, **Innovation**, **Ownership**, **Analytical Skills**, **Communication Skills**, **Time Management**, **Self-driven**, **Flexible**